

Bridging Spatial and Social Network Analysis Communities through Visual Analytics for Collaborative Work: A Case Study

Sichen Jin

School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
sjin86@gatech.edu

Alex Endert

School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
endert@gatech.edu

Ben Rydal Shapiro

Department of Learning Sciences
Georgia State University
Atlanta, Georgia, USA
bshapiro@gsu.edu

Clio Andris

School of City & Regional Planning and Interactive
Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
clio@gatech.edu

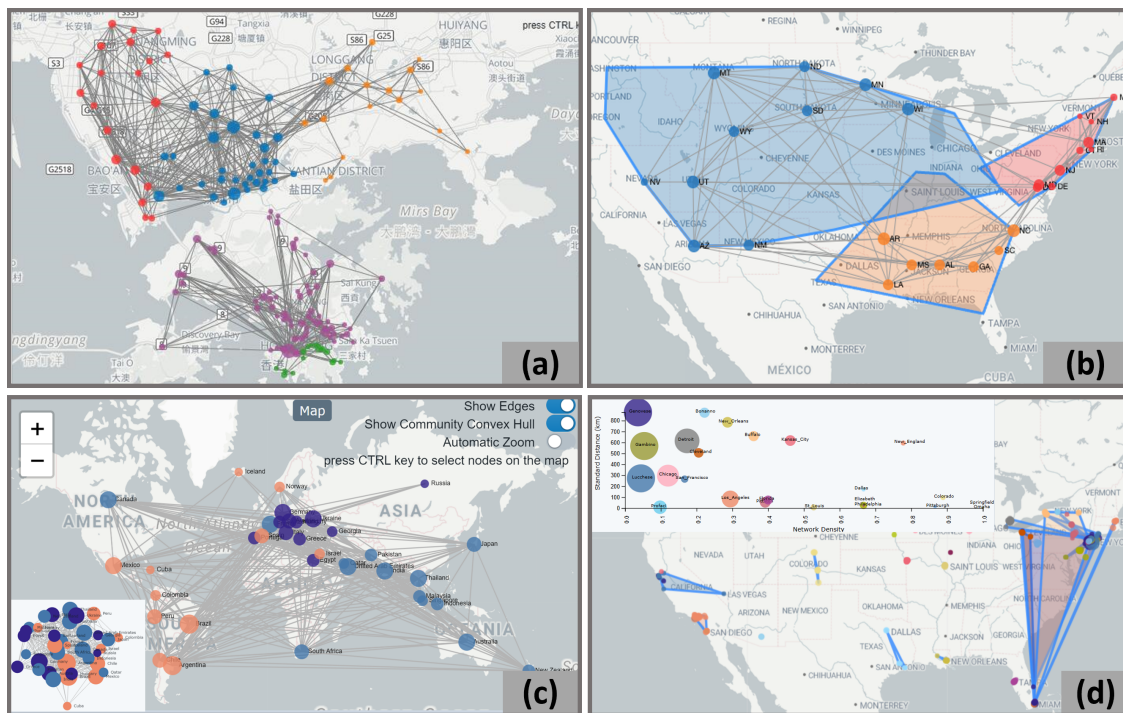


Figure 1: Data insights shared by participants during the workshop: (a) a commuting network in southern China, (b) a U.S. migration network with convex hulls outlining spatial expanses of network communities, (c) an international wheat trade network, and (d) an American Mafia family criminal network with a Cluster-Cluster plot.

Abstract

For years, geographic information systems (GIS) and social network analysis (SNA) have evolved as distinct research communities with their own tools, vocabulary, and infrastructure. However, there remains a lack of approaches for integrating SNA and GIS research to model interpersonal relationships in geographic space and advance topics central to HCI, such as investigating criminal associations and designing location-based services. This paper presents a

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workshop involving GIS and SNA professionals, centered around SNoMaN, a novel tool for spatial social network analysis. We use qualitative methods to examine the practices and insights that emerged as participants considered integrated approaches across SNA and GIS. We detail the impact of participants' backgrounds on their analytical perspectives and tool usage preferences and identify points of friction, similarities, and differences in language and data use. Our findings highlight the advantages and implications of using visual analytics to support interdisciplinary work.

CCS Concepts

• **Human-centered computing** → **Visual analytics**; **Empirical studies in visualization**; *Geographic visualization*; **Empirical studies in collaborative and social computing**.

Keywords

social network analysis, geographic information science, interdisciplinary workshop, visual analytics

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1 Introduction

For years, geographic information systems (GIS) and social network analysis (SNA) have evolved as distinct communities each with their own tools, vocabulary, and infrastructure that continue to influence HCI researchers and designers in a variety of ways. GIS focuses on the spatial aspects of data, providing extensive ways to store, process, and visualize geographic information. It uses specialized software such as ArcGIS [21], and QGIS [36] and relies heavily on geostatistics, cartographic principles, etc. In contrast, SNA focuses on understanding relationships within networks, using tools such as Gephi [6], Pajek [7], and UCINET [11] as well as methods such as centrality measures and community detection algorithms to analyze the structure and dynamics of social networks.

Despite these differences, GIS and SNA share many analogies, methods, and interests. For example, the likelihood for social interaction decreases with geographic distance [29, 32] and landscapes and geographical boundaries can also play a role. Likewise, existing research highlights how waterways guide social group formations between Amazonian households [2], land use impacts tractor hiring patterns in rural Thailand [22], and about 75% of Twitter connections are within the same country [42]. These phenomena illustrate how geographic factors and social connections are deeply intertwined in many real-world scenarios. Social networks are often influenced by geographic constraints, while spatial organization is, in turn, shaped by social behaviors. Conducting joint GIS and SNA research enables a more comprehensive understanding of these interplays between spatial and social spaces.

However, little research has developed methods for interdisciplinary collaboration between GIS and SNA in ways that can

support HCI researchers in a variety of areas. Notably, such collaboration is crucial for advancing the analysis of spatial social networks, which represent interpersonal or inter-organizational relationships with nodes and edges having defined geographic locations. Spatial social network analysis (SSNA) has applications in and outside of HCI, including investigating criminal associations [3], providing location-based services [14, 29], epidemiology [23, 27, 39, 43], and studying telecommunication patterns [9, 38, 46]. Importantly, integrating GIS and SNA poses challenges due to the differences in professional training and methodological and conceptual frameworks between SNA and GIS.

To bridge this gap and unpack the challenges and needs for interdisciplinary studies across SNA and GIS, we present a case study of a design workshop involving professionals and students from both fields, centered around SNoMaN, a visual analytic tool for spatial social network analysis that combines methods from both GIS and SNA to provide more integrated workflows. For example, it enables the calculation and comparison of metrics from both fields, establishes a shared data format, and uses visualizations and interactions to mitigate barriers and smooth transitions between geographic and network spaces. We used SNoMaN as a probe to mediate and foster knowledge exchange and identify barriers to collaborative work in these communities. We employed qualitative methods to analyze the language, ideas, and practices that emerged, focusing on the following research questions:

RQ1 What new insights did SNA and GIS researchers generate when studying spatial social networks together with a combination of GIS and SNA methods?

RQ2 How do participants characterize values, applications, and challenges when integrating spatial and social network communities through visual analytics?

Our findings highlight the potential of visual analytics as a boundary object [12] to facilitate interdisciplinary work around data across GIS and SNA communities. Specifically, we show how visual analytic tools provide integrated environments for exploratory data analysis that enable participants to examine diverse aspects of datasets, fostering interdisciplinary learning. We also show how these tools serve as aesthetic provocations [4, 25, 37] for participants to think critically and challenge disciplinary preconceptions. Additionally, we identify "science friction" [19, 20, 34] between SNA and GIS, such as differing analytical perspectives and variations in terminology, offering insights into interdisciplinary collaboration. Our work underscores both the challenges and opportunities of using visual analytics to bridge interdisciplinary communities such as SNA and GIS to support work in and outside of HCI.

2 Methods

2.1 Workshop Design and Procedure

We hosted a workshop as the empirical basis of this work, with the intent of facilitating *mutual learning* among the dual disciplinary participants and the tool designers and allowing participants to "design by doing" [30] with the SNoMaN tool. Twenty-six GIS and SNA researchers, including professors, graduate students, and practitioners from diverse fields such as supply chain, crisis management, human mobility, epidemiology, computational social science, and geospatial economics, voluntarily participated (Figure 2). The study



Figure 2: Participants were grouped into teams of 5 to 6 individuals and seated at round tables in a conference room. The picture was taken while participants were using the tool to complete the lab exercise. Participants' faces are blurred due to privacy concerns.

received approval from the institutional review board (IRB), ensuring ethical standards were maintained throughout the user study and included six parts:

(1) *Introductory Tutorial.* As participants came with various background levels in GIS and SNA, we began with introductory tutorials (approximately 15 minutes) of the SNoMaN tool to help participants learn about the tool functionalities (Figure 3) and review GIS and SNA concepts and methods.

(2) *Initial Data Exploration.* Participants then worked collaboratively to explore a US flight network and complete SSNA exercises (approximately 30 minutes). Example exercises include importing datasets, interacting with data through selections/filters/highlights, configuring visualizations, computing network statistics, identifying spatial and social correlations and outliers, and detecting and outlining network modules and their spatial expanses on the map.

(3) *Additional and Personal Data Exploration.* Subsequently, participants were provided three other preloaded example SSN datasets to explore, including a Mafia SSN [16], a food sharing SSN [18], a U.S. congressional voting SSN [28]. Participants were also encouraged to load their own datasets (e.g., from their own work contexts) into the tool to explore how SSNA and the SNoMaN tool might support their own work (approximately 30 minutes).

(4) *Collaborative Discussion.* A 30-minute participant-driven discussion allowed groups to share their experiences, problems, and suggestions for the SNoMaN tool and SSNA. We audio-recorded and took notes during the discussion.

(5) *Participant Presentations.* Several participants presented visualizations and insights generated using SNoMaN on their datasets. These presentations sparked discussion and highlighted diverse applications (approximately 30 minutes).

(6) *Workshop Survey.* Finally, participants were encouraged to complete a survey (approximately 15 minutes). It assessed their research backgrounds, tool usage experience, and findings through Likert-scale questions and open-ended responses. Participants were also encouraged to upload screenshots and notes documenting their generated insights.

2.2 SNoMaN, A Visual Analytic Tool for Spatial Social Network Mapping and Analysis

SNoMaN is a visual analytic tool designed to support the mapping and analysis of spatial social networks. At a high level, it links a sociogram [5] (Figure 3 (a) A) with a geographic network layout (Figure 3 (a) B) using brushing and linking interactions. The sociogram is a classic presentation in SNA that reveals the overall topological structures of a network and aids in identifying topologically significant nodes or groups, such as central nodes with many connections or bridge nodes connecting different groups. On the other hand, displaying the network's geographic layout on a basemap allows users to observe the spatial distribution of nodes and connections, which connect these network elements to their geographic surroundings within the landscape (e.g., edges crossing rivers, and nodes in isolated areas). The tool, therefore, offers a side-by-side comparison of node placements in both a social and geographic context, enhancing participants' ability to assess the mutual influence between the two dimensions.

Additionally, the software allows participants to interactively compute and visualize GIS and SNA metrics [35]. These metrics include network density, network diameter, clustering coefficient, edge Euclidean distance distribution, and node degree distribution (Figure 3 (a) C). The software also calculates node centrality metrics such as degree, betweenness, and closeness centralities. Moreover, participants can run a community detection algorithm (specifically, the Louvain algorithm [10]) within the tool to partition the network into densely connected groups. The tool then visually represents these groups on the map in different colors. Participants can outline the geographic expanses of network groups using convex hulls. Participants are also given the ability to adjust nodes' visual properties (e.g., size and color) or filter nodes based on these computed metrics or attributes loaded with the dataset (Figure 3 (a) E).

Furthermore, the software provides an interactive scatterplot that allows participants to explore the relationships between GIS and SNA metrics (Figure 3 (a) D). For example, participants can investigate how a node's number of connections correlates with its distance from the network center by plotting one metric against the other. Moreover, participants can compute and compare the length of shortest path and Euclidean distance between pairs of nodes to

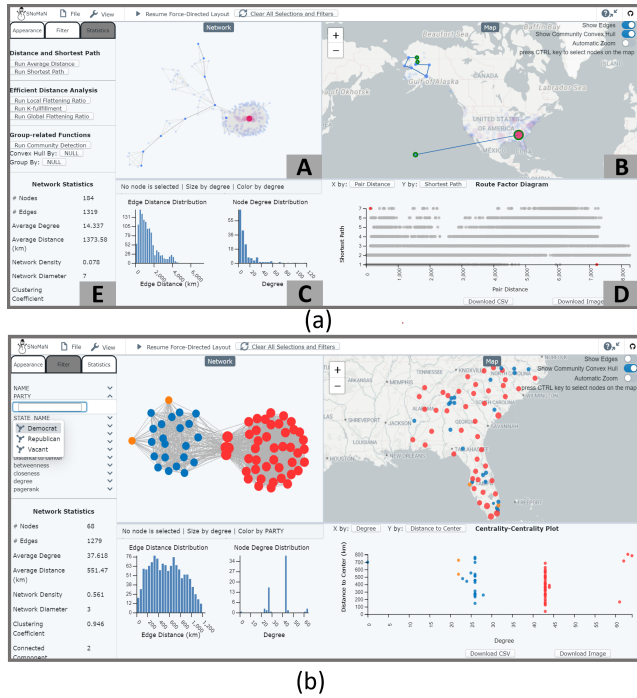


Figure 3: The SNoMaN user interface showcases two preloaded example datasets: (a) a US flight network [33] and (b) a congressman voting SSN [28]. The tool consists of (A) a network view presenting the force-directed layout (i.e., sociogram) of the network, (B) a map view presenting the geographic layout of the network, (C) two histograms dynamically displaying edge Euclidean distance and node degree distribution of users' selection, (D) a scatterplot examining the relationships between SNA and GIS metrics, and (E) a configuration and function menu that supports configuring network appearance, filtering by nodes attributes, running analysis, and computing network statistics dynamically. Brushing and linking interactions are supported across all views.

identify any correlations or outliers in the network. Participants can also calculate and compare group-level metrics like standard distance (measuring the spatial dispersion of group members) and group network density (measuring the connectivity within a group) to test if spatially clustered groups tend to have denser connections. To enhance the exploratory data analysis experience, all views are interconnected through brushing and linking interactions. This means that participants can select nodes, paths, or groups of interest in one view to highlight and retrieve additional information about them in the other linked views. The tool and its visualization and interaction features provide opportunities for participants to observe and comprehend the interplay between social network and geographic spaces.

2.3 Data and Analysis

We documented workshop activities using audio/video recordings, photographs, and participant-created artifacts. Additionally, we gathered completed survey responses from 15 participants. Due to the survey's anonymity, we could not match these responses with individuals from the audio or video recordings. While only 15 respondents completed the survey, these responses provide valuable data to answer our research questions and a window into the diverse GIS/SNA backgrounds of the larger group. The 15 respondents, numbered P1 to P15 (Figure 4), included one associate professor, two postdoctoral researchers, two research scientists, nine graduate students, and one undergraduate student. Most were engaged in SSNA-relevant fields, with a median of five years and an average of seven years in their respective research areas. Participants rated their expertise in SNA and GIS on a scale of 0 to 10, where 10 indicated significant expertise (e.g., publishing papers or completing coursework) and 0 indicated no prior exposure (Figure 4).

After transcribing audio and video recordings, we conducted qualitative analysis in the grounded theory tradition, triangulating across our data sources [13, 15, 41]. Using constant comparison, we iteratively developed and refined emergent themes. The research team met bi-weekly throughout this process for a period of six months, maintaining a codebook with definitions and examples of emerging conceptual categories and resolving disagreements to further refine codes and conceptual categories. This process led to the following three primary themes.

3 Findings

3.1 Tool Usage Preferences Vary by GIS and SNA Expertise Level

The first emergent theme was the influence of participants' GIS and SNA backgrounds on their perceptions of the SNoMaN tool. We developed this theme exclusively from the survey responses of 15 participants. Figure 5 presents five scatterplots of Likert scale questions reflecting participants' reactions to the SNoMaN tool. Each dot represents one of the 15 respondents (P9 and P14 are jittered for clarity), with the y-axis indicating self-assessed SNA expertise and the x-axis showing GIS expertise. Dot colors correspond to the 5-point Likert scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree, Not Sure).

Overall, GIS researchers tended to express stronger agreement about the tool's usefulness for research questions and classroom use, with those choosing "strongly agree" having extensive GIS expertise. In contrast, participants with relatively low SNA and GIS expertise were less positive about the tool's usage in the classroom, for answering research questions, generating new ideas, and ease of use. For instance, while most participants understood the tool's terminology, three (P4, P9, P14) participants expressed difficulty in understanding the concepts, all of whom scored SNA expertise below four.

Participants with notable differences in SNA and GIS expertise (P4, P8, P9, P13, P15) often "strongly agreed" that the tool helped generate new ideas about spatial networks. For instance, P15 (SNA 8, GIS 2) noted that "many network measures that typically signal social or interaction closeness can also and simultaneously be tested



Figure 4: The SNA and GIS expertise level distribution among participants.

in a spatial way.” Meanwhile, P4 (SNA 1, GIS 10) found insights from exceptions where connectivity does not align with spatial proximity, such as distant airports being connected (e.g., Atlanta and Hawaii) or nearby airports being separated (e.g., a chain in Alaska), challenging the assumption that “near things are more related than distant things” [44].

3.2 Generating New Insights and Questions about the Interaction Between GIS and SNA

Our second theme highlights specific insights and questions raised by participants about the interaction between GIS and SNA, drawing from survey responses and workshop recordings. 7 out of 15 participants strongly agreed, and 8 out of 15 participants agreed that SNoMaN helped them generate new ideas about spatial networks (Figure 5). Four participants (P3, P9, P11, P14) mentioned the tool made them rethink their research problems and would apply these new perspectives to their future research.

Open-ended survey responses provided further detail. For example, P11 noted, “SNoMaN makes it easy to discover, like, not just things about your data, but things about like spatial social network analysis itself, like concepts. You can actually sort of feel yourself learning about it.” Likewise, P10 stated, “It is very useful to add spatial context to network distance, community detection, and other forms of SNA. It led to some aha moments.” Similarly, participants exploring the US airports noted unexpected patterns: “[There are] chain patterns in Alaska where two airports are geographically close but separated by several hops in the network” (P7); and “it was surprising to see how low the network density was and that geographic distance did not seem to indicate the probability of a link.” (P10).

Participants also shared insights from using SNoMaN on their own datasets. One participant illustrated insights from a commuting flow network in Hong Kong and Shenzhen, noting, “In the specific case of Shenzhen, the communities were primarily divided by mountains, whereas in Hong Kong, two communities were found to be separated by the harbor” (Figure 1a). Another demonstrated use cases on an international wheat import SSN, identifying a spatially close and densely connected group in Eastern Europe, which turned out to involve Russia and Ukraine (Figure 1c). One participant successfully replicated her research findings on US migration SSNs using community detection functions, revealing spatial clustering of immigration flows (Figure 1b). These insights spurred discussions on reasons behind these patterns, such as travel costs and cultural differences.

In addition, a researcher with a background in sociology used convex hulls to outline the activity areas of Mafia families in a criminal social network (Figure 1d). The visualization revealed that most families are geographically clustered, operating within the same city or nearby areas. He proposed a sociological hypothesis for unexpected geographic distributions, noting that *the only real exception is Miami, which is like an open city that any family could send representatives to, usually to set up gambling operations.* He added that some families maintained long-distance ties for business reasons, such as the drug trade, which *requires moving things across space that may drive these cross-coast communities.* He reflected, “even after looking at this data for almost ten years, I was actually able to see patterns that I hadn’t noticed before by examining the network in this spatial context side by side.”

Collectively, these responses highlight how GIS and SNA integration helped participants generate insights, spark cross-disciplinary discussions, and facilitate understanding the interplay between spatial and social spaces, such as identifying disconnected geographic neighbors, recognizing geographic segregation, measuring spatial dispersion, evaluating correlations between connection frequency and geographic proximity and nodes’ social roles with locations.

3.3 Challenges and Advantages with the Interaction Between GIS and SNA

While participants highlighted many insights and new research questions from the interaction between GIS and SNA, they also identified challenges in collaborative work across these communities. The first challenge is the difficulty in comprehending unfamiliar terms and metrics from the other discipline. Specifically, three participants (P4, P9, P14) primarily skilled in GIS faced challenges in understanding network terms and concepts. For instance, P4 stated that “It was challenging to understand the community detection algorithm” and P9 suggested that “Some of the terms were not immediately clear to me. It could be helpful to incorporate pop-ups with definitions.”

The second challenge we observed involves data compatibility issues across SNA and GIS tools. This challenge was discussed and raised by several workshop participants. The data formats supported by SNA tools (e.g., Gephi) usually require an edge list file specifying connections between nodes and an optional node list file with nodes’ attributes, including nodes’ geographic location information. Conversely, GIS tools often use formats like GeoJSON or Shapefile to represent nodes and edges as geometries or vectors

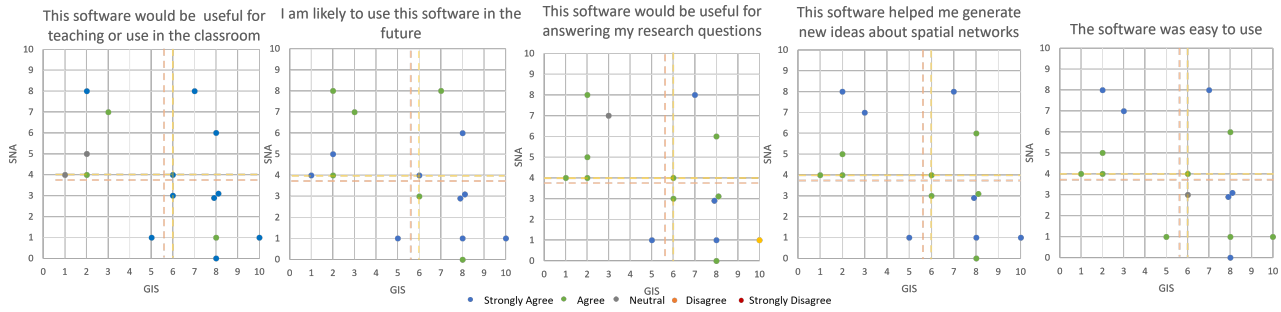


Figure 5: The Likert evaluation results of participants in relation to their research expertise in social network analysis and geographic information science.

with coordinates. The data format incompatibility requires manually translating data by constantly switching between separate tools, which can hinder the fluidity of exploratory research.

On the other hand, participants highlighted the integration of network and geographic visual analytics in SNoMaN as a key advantage. For example, P15 stated, *“The most helpful panel for me is the parallel view as a social network and then again plotted on a geographical map. It allowed for directly comparing social and spatial distance, social node attributes and geographical ones.”* P9 added, *“I really liked the multiple views at the same time. [...] Its capability of approaching social networks with a geographic perspective is what other tools don’t do.”* During discussions, participants noted the novelty of examining network structures and spatial relationships simultaneously, with one participant stating that *“[This might be an] obvious thing, but being able to look at both the network structure and look at it on a map and see it spatially at the same time is helpful. You don’t really have that luxury in having two tools open at the same time.”* Another participant agreed, stating that *“I have to have two tools open at the same time for my work right now so that I can compare. I found that this was one of the most integrated tools that I’ve seen regarding its ability to capture all the different ways that you need to be thinking about networks in a very intuitive kind of accessible way.”*

Participants also liked the interaction-enhanced exploratory process supported by SNoMaN. One participant stated, *“I appreciate the ability to draw a box and then see specific nodes in the network and related statistics and histograms.”* P14 elaborated on his data insights generation process, stating, *“By utilizing the filter tool, I was able to conveniently remove aggregated spots of little significance, leading to the identification of five distinct communities using the statistics tool.”* P9 highlighted the tool’s analytical potential with high interactivity: *“The whole concept of seeing geography and the network at the same time is awesome—not just for visualization but for relating metrics from both.”* Additionally, three participants (P3, P5, P15) acknowledged SNoMaN’s advantages in *“quick visual interactivity, intuitive and easy to pick up (no need to install any package or app), straightforward”,* which makes it better for *“learning”, “in-the-field analysis”, “quick visualization and exploration”, and “providing policy advice for policymakers”*. These comments demonstrate the crucial role of interaction in exploratory spatial data analysis. Interactive

features in visual analytics, such as brushing and linking and filtering, help maintain diverse contexts and continuity in participants’ analyses and allow users to dynamically connect different data aspects, facilitating rapid investigation of identified relationships and patterns.

However, the multiview design also posed challenges in navigation and view coordination. With multiple views providing different contextual presentations, the same nodes appear in social, geographic, and feature spaces. P9 commented, *“The navigation can be a bit difficult, and the selection was counterintuitive sometimes.”* Conflicting interaction conventions between SNA and GIS tools were also noted. For example, in sociograms, “click and drag” typically selects a node, whereas in map views, it pans the map. Participants found these differences confusing, emphasizing the need for future designs to enhance navigation and interaction consistency in SNoMaN and similar cross-disciplinary tools.

4 Discussion

4.1 Challenges and Opportunities for GIS and SNA Interdisciplinary Studies

The notion of “science friction” refers to tensions that arise in interdisciplinary collaboration, including practical (e.g., data format conversion) and conceptual challenges (e.g., differences in terminology usage) [19, 20, 34]. In the context of combining GIS and SNA, we have identified several science frictions and strategies to potentially address them.

Most notably, the lack of a common language and differing interpretations of shared terms (e.g., “distance”, “clusters”, “adjacency”) poses challenges in tool usage and cross-community communication. This challenge of jargon is well recognized in multidisciplinary collaborations [26, 31]. Our work highlights how tools such as SNoMaN address these major types of friction identified in prior interdisciplinary work [19, 20] by standardizing data formats, incorporating both geographic and network information into unified files, reducing time and effort for data process.

At the same time, science friction can also create opportunities for interdisciplinary innovation, fostering new insights and hypotheses [34]. During our workshop, GIS and SNA participants contributed diverse perspectives, providing a holistic understanding of spatial social networks. Questions raised included: “How does

geographic closeness affect network density?” “How do network paths translate in the territory?” “Do network clusters align with administrative/linguistic/geographic boundaries?” In addition, while the geographic influences on tie formation were often more evident, social network analysts also contributed sociological explanations for geographic patterns (Section 3.2). This exchange of methods and sensemaking fostered the integration of perspectives from the other domain, yielding new ideas and prompting participants to rethink their research questions. The diversity in participants’ disciplinary backgrounds proved crucial for generating innovative insights, including unexplored connections between geography and social spaces. The findings also suggest that tools supporting multidisciplinary collaboration should account for these interdisciplinary dynamics by considering team members’ backgrounds and expertise [1, 45].

4.2 Visual Analytics As a Boundary Object for Collaborative Work Across GIS and SNA Communities

The concept of a boundary object is crucial in collaborative work between communities of practice [12]. Boundary objects are “objects that both inhabit several communities of practice and satisfy the informational requirements of each of them” [40]. Bhavnani et al. [8] suggested that using visual analytics as a boundary object “can assist in integrating diverse disciplinary knowledge and enable the team to arrive at novel translational insight” [8]. We found that SNoMaN served as a type of boundary object to facilitate collaborative work across GIS and SNA. In the following, we first discuss why this was the case, and then expand upon the value of visual analytics as a means to encourage cross-disciplinary collaboration.

SNoMaN demonstrated the following steps towards integration across GIS and SNA communities. First, it allows for calculations and comparisons between SNA and GIS measurements of node, path, and group for correlation and outlier analysis. Second, the tool set up an initial data format standard for SSNA. Third, it introduces a new workflow that combines SNA and GIS methods (e.g., using convex hulls to visualize social group activity areas). Fourth, it uses visualizations and interactions to smooth data transitions and representations between geographic and network spaces.

More broadly, this work extends the literature on visual analytics as boundary objects to support interdisciplinary collaboration. In particular, our analysis highlights how visualization helped participants comprehend metrics from unfamiliar fields, promoting interdisciplinary learning [17]. For example, participants used visualizations to present and discuss GIS and SNA concepts, such as outlining group areas with convex hulls (Section 3.2). SNoMaN also served as an aesthetic provocation [4, 25, 37] for participants to engage in critical thinking and challenge preconceived notions within their disciplinary boundaries, such as how the Alaska airport chain pattern provokes participants to rethink the assumed correlation between geographic distance and connectivity. In addition, our work demonstrates the essential role of interaction in supporting cross-disciplinary work. Fluid interactivity and techniques like brushing and linking integrated different views of data [24], helped participants dynamically calculate metrics cross disciplines, and

facilitated the identification of correlations and outliers together without needing to manually transferring data between tools.

5 Limitations and Future Work

This work has several limitations. First, the study is based on one design workshop and the number of participants was relatively small. Although we intentionally recruited diverse perspectives spanning SNA and GIS, the workshop and analysis may not capture the full range of practices. Second, while we recorded open-ended discussions and attempted to link audio recordings with survey data, this was challenging as participants often speaking simultaneously. Additionally, the tool may not have been perfect and could have contained bugs that impacted the user experience. Future work will include multiple workshops with varied participant groups and using a mix of qualitative and quantitative methods for data collection and analysis. Conducting longitudinal studies to further explore how interdisciplinary collaborations across GIS and SNA communities develop and how tools like SNoMaN are adopted in real-world settings would also provide deeper insights about these communities and the effectiveness of visual analytics to support collaborative work.

6 Conclusion

Analyzing SSNs requires expertise in geographic information systems and social network analysis, yet little research has focused on supporting integrated analysis across these domains. Many research endeavors, including analyzing online and offline user behavior, monitoring human mobility, investigating criminal associations, modeling epidemics, etc. are best conducted with joint network and geospatial tools. This work used a visual analytics tool, SNoMaN, to explore challenges and opportunities in interdisciplinary collaboration among GIS and SNA researchers in an interdisciplinary workshop. Our findings demonstrated how participants’ backgrounds shape their analytical perspectives and tool usage preferences. We also identified science frictions, such as the lack of a common language and inconsistent data formats. The study highlights the potential of using visual analytics as a bridge between GIS and SNA communities, offering insights for designing tools that support integrative exploratory analysis in interdisciplinary work particularly relevant to HCI researchers and designers.

References

- [1] Piotr D Adamczyk and Michael B Twidale. 2007. Supporting multidisciplinary collaboration: requirements from novel HCI education. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 1073–1076.
- [2] Clio Andris. 2016. Integrating social network data into GISystems. *International Journal of Geographical Information Science* 30, 10 (2016), 2009–2031.
- [3] Clio Andris, Daniel DellaPosta, Brittany N Freelin, Xi Zhu, Bradley Hinger, and Hanzhou Chen. 2021. To racketeer among neighbors: Spatial features of criminal collaboration in the American Mafia. *International Journal of Geographical Information Science* 35, 12 (2021), 2463–2488.
- [4] Dagny Aurich and Aida Horaniet Ibañez. 2023. How can data visualization support interdisciplinary research? LuxTIME: Studying historical exposomics in Belval. *Frontiers in big Data* 6 (2023).
- [5] Michael J Bannister, David Eppstein, Michael T Goodrich, and Lowell Trott. 2013. Force-directed graph drawing using social gravity and scaling. In *Graph Drawing: 20th International Symposium, GD 2012, Redmond, WA, USA, September 19–21, 2012, Revised Selected Papers 20*. Springer, 414–425.
- [6] Mathieu Bastian, Sebastien Heymann, and Mathieu Jacomy. 2009. Gephi: An open source software for exploring and manipulating networks. In *Third International AAAI Conference on Weblogs and Social Media*.

- [7] Vladimir Batagelj and Andrej Mrvar. 2004. Pajek—analysis and visualization of large networks. In *Graph drawing software*. Springer, 77–103.
- [8] Suresh K Bhavnani, Shyam Visweswaran, Rohit Divekar, and Allan R Brasier. 2019. Towards team-centered informatics: accelerating innovation in multidisciplinary scientific teams through visual analytics. *The Journal of Applied Behavioral Science* 55, 1 (2019), 50–72.
- [9] Vincent Blondel, Gautier Krings, and Isabelle Thomas. 2010. Regions and borders of mobile telephony in Belgium and in the Brussels metropolitan zone. *Brussels Studies. La revue scientifique pour les recherches sur Bruxelles/Het wetenschappelijk tijdschrift voor onderzoek over Brussel/The Journal of Research on Brussels* (2010).
- [10] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefevre. 2008. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* 2008, 10 (2008), P10008.
- [11] Stephen P Borgatti, Martin G Everett, and Linton C Freeman. 2002. Ucinet for Windows: Software for social network analysis. *Harvard, MA: Analytic Technologies* 6 (2002), 12–15.
- [12] Geoffrey C Bowker and Susan Leigh Star. 2000. *Sorting things out: Classification and its consequences*. MIT press.
- [13] K. Charmaz. 2014. *Constructing grounded theory* (2nd edition ed.). Sage, Los Angeles, CA.
- [14] Eunjoon Cho, Seth A Myers, and Jure Leskovec. 2011. Friendship and mobility: user movement in location-based social networks. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*. 1082–1090.
- [15] Juliet Corbin and Anselm Strauss. 2014. *Basics of qualitative research: Techniques and procedures for developing grounded theory*. Sage publications.
- [16] Daniel DellaPosta. 2017. Network closure and integration in the mid-20th century American mafia. *Social Networks* 51 (2017), 148–157.
- [17] Anna Dyson. 2016. Interactive visualization for interdisciplinary research. *Visualization and Data Analysis*.
- [18] Jaimie Elizabeth Edwards. 2020. *Over the river and through the woods: Examining the relationship between network structure, collaboration and geography*. Ph.D. Dissertation. Virginia Tech.
- [19] Paul N Edwards. 2013. *A vast machine: Computer models, climate data, and the politics of global warming*. MIT press.
- [20] Paul N Edwards, Matthew S Mayernik, Archer L Batcheller, Geoffrey C Bowker, and Christine L Borgman. 2011. Science friction: Data, metadata, and collaboration. *Social studies of science* 41, 5 (2011), 667–690.
- [21] ESRI. 2011. *ArcGIS Desktop: Release 10*. Redlands, CA.
- [22] Katherine Faust, Barbara Entwisle, Ronald R Rindfuss, Stephen J Walsh, and Yothin Sawangdee. 2000. Spatial arrangement of social and economic networks among villages in Nang Rong District, Thailand. *Social networks* 21, 4 (2000), 311–337.
- [23] Song Gao, Krzysztof Janowicz, and Helen Couclelis. 2017. Extracting urban functional regions from points of interest and human activities on location-based social networks. *Transactions in GIS* 21, 3 (2017), 446–467.
- [24] Jeffrey Heer and Ben Shneiderman. 2012. Interactive dynamics for visual analysis: A taxonomy of tools that support the fluent and flexible use of visualizations. *Queue* 10, 2 (2012), 30–55.
- [25] Uta Hinrichs, Stefania Forlini, and Bridget Moynihan. 2019. In defense of sandcastles: Research thinking through visualization in digital humanities. *Digital Scholarship in the Humanities* 34, Supplement_1 (2019), i80–i99.
- [26] Maaïke Kleinsmann and Rianne Valkenburg. 2008. Barriers and enablers for creating shared understanding in co-design projects. *Design studies* 29, 4 (2008), 369–386.
- [27] Minh Tri Le, David Attaway, Taylor Anderson, Hamdi Kavak, Amira Roess, and Andreas Züfle. 2022. PhyloView: A system to visualize the ecology of infectious diseases using phylogenetic data. In *2022 23rd IEEE International Conference on Mobile Data Management (MDM)*. IEEE, 222–229.
- [28] Jeffrey B. Lewis, Keith Poole, Howard Rosenthal, Adam Boche, Aaron Rudkin, and Luke Sonnet. 2024. Voteview: Congressional Roll-Call Votes Database. <https://voteview.com/>
- [29] Yiding Liu, Tuan-Anh Nguyen Pham, Gao Cong, and Quan Yuan. 2017. An experimental evaluation of point-of-interest recommendation in location-based social networks. *Proceedings of the VLDB Endowment* 10, 10 (2017), 1010–1021.
- [30] Rachael Luck. 2018. What is it that makes participation in design participatory design? *Design studies* 59 (2018), 1–8.
- [31] Mimi Nguyen and Céline Mougenot. 2020. Multidisciplinary design collaboration in the lenses of CSCW and current technology enablement. (2020).
- [32] Jukka-Pekka Onnela, Samuel Arbesman, Marta C González, Albert-László Barabási, and Nicholas A Christakis. 2011. Geographic constraints on social network groups. *PLOS One* 6, 4 (2011), e16939.
- [33] OurAirport. 2017. Airport, airline and route data. <https://ourairports.com/data/>
- [34] Georgia Panagiotidou, Jeroen Poblome, Jan Aerts, and Andrew Vande Moere. 2022. Designing a Data Visualisation for Interdisciplinary Scientists. How to Transparently Convey Data Frictions? *Computer Supported Cooperative Work (CSCW)* 31, 4 (2022), 633–667.
- [35] Christina Prell. 2011. Social network analysis: History, theory and methodology. *Social Network Analysis* (2011), 1–272.
- [36] QGIS Development Team. 2024. *QGIS Geographic Information System*. QGIS Association. <https://www.qgis.org>
- [37] Dimitrios Raptis, Rikke Hagensby Jensen, Jesper Kjeldskov, and Mikael B Skov. 2017. Aesthetic, functional and conceptual provocation in research through design. In *Proceedings of the 2017 Conference on Designing Interactive Systems*. 29–41.
- [38] Markus Schläpfer, Luis MA Bettencourt, Sébastien Grauwin, Mathias Raschke, Rob Claxton, Zbigniew Smoreda, Geoffrey B West, and Carlo Ratti. 2014. The scaling of human interactions with city size. *Journal of the Royal Society Interface* 11, 98 (2014), 20130789.
- [39] Yuri P Springer, Michael C Samuel, and Gail Bolan. 2010. Socioeconomic gradients in sexually transmitted diseases: A geographic information system-based analysis of poverty, race/ethnicity, and gonorrhea rates in California, 2004–2006. *American journal of public health* 100, 6 (2010), 1060–1067.
- [40] Susan Leigh Star and James R Griesemer. 1989. Institutional ecology, translations' and boundary objects: Amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39. *Social studies of science* 19, 3 (1989), 387–420.
- [41] A. Strauss and J.M. Corbin. 1997. *Grounded theory in practice*. Sage, Los Angeles, CA.
- [42] Yuri Taktcheyev, Anatoliy Gruz, and Barry Wellman. 2012. Geography of Twitter networks. *Social Networks* 34, 1 (2012), 73–81.
- [43] Alok Tiwari and Mohammed Aljoufie. 2021. A qualitative geographical information system interpretation of mobility and COVID-19 pandemic intersection in Uttar Pradesh, India. *Geospatial Health* 16, 1 (2021).
- [44] Waldo R Tobler. 1970. A computer movie simulating urban growth in the Detroit region. *Economic geography* 46, sup1 (1970), 234–240.
- [45] Maria G Wadlow. 1993. Visual Interaction Design Special Interest Area: Interdisciplinary Design. *ACM SIGCHI Bulletin* 25, 3 (1993), 71–74.
- [46] Yaoli Wang, Chaogui Kang, Luis MA Bettencourt, Yu Liu, and Clio Andris. 2015. Linked activity spaces: Embedding social networks in urban space. *Computational approaches for urban environments* (2015), 313–336.